

Total No. of Questions : 4]

SEAT No. :

PD455

[6409]-244R

[Total No. of Pages : 1

S. E. (Artificial Intelligence and Machine Learning) (Insem)

COMPUTER GRAPHICS

(2019 Pattern) (Semester-IV) (218555)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2 Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Classify the major groups of graphics function in openGL. Explain in detail with suitable example. [8]

b) What is computer graphics? Explain the concept of display file structure. [7]

OR

Q2) a) Consider a line P (6, 6) to Q (15, 12). Use Bresenham's line drawing algorithm to rasterize the line from P to Q. Draw the pixel wise rasterization of Line. [8]

b) Explain with diagram the difference between raster scan and random scan displays. [7]

Q3) a) What are the steps involved in scan line polygon filling algorithm. [8]

b) Perform a 60 degree rotation of triangle A (0, 0), B (30, 60), C (60, 0) in anticlockwise direction. [7]

i) About the origin (0, 0)

ii) About a point P (30, 03)

OR

Q4) a) Consider the square A (0,0), B (150,0), C (150,150), D (0,150). Rotate the square by 60 degree anti-clockwise direction followed by reflection about Y-axis. [8]

b) Show that the composition of two rotations is additive. [7]

$$R(\Theta_1).R(\Theta_2) = R(\Theta_1 + \Theta_2)$$

